

# Nikhil Mehta

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## RESEARCH INTERESTS

Deep Learning, Machine Learning, Probabilistic Models, Optimization

## EDUCATION

**Delhi Technological University**, New Delhi, India 2010-2014  
*B.Tech.* in Computer Software Engineering

## RESEARCH EXPERIENCE

**Generative models and their applications** July '17 - Present  
*Guided by Prof. Piyush Rai at IIT Kanpur*

- Proposed a *novel* Bayesian nonparametric deep generative framework to model network data for overlapping community detection and membership strength. The framework allowed fast inference over previously used interpretable stochastic block models which used MCMC methods.
- Implemented an extension of the vanilla recurrent variational auto-encoder. The architecture had a combination of encoder-generator and a discriminator (trained in wake-sleep fashion) that allowed the network to learn structured disentangled representations. Discriminator performance for text classification improved by 3-4% with only 3000 labeled examples. [\[code\]](#)
- Implemented a wasserstein generative-adversarial-network (W-GAN) to synthesize data for unseen classes conditioned on class attributes. Preliminary results suggest a competitive zero-shot classification accuracy on Caltech-UCSD-Bird Dataset. [\[code\]](#)

**Learning based Localization of Autonomous Vehicles** Mar - Jun '17  
*Guided by Prof. Gaurav Pandey at IIT Kanpur*

- Created localization datasets (based on ground-reflectivity maps) from 3D lidar scans for machine learning tasks such as 1) Pose-Regression and 2) Metric Learning.
- Implemented a CNN based discriminative metric model to learn invariant representations which were used to calculate similarity scores between prior and local maps.
- Implemented a capsule based architecture to learn equivariant representations which could be used for pose regression. Open-sourced a proof of concept implementation of “Transforming Autoencoders” to learn transformations. [\[code\]](#)

**MI based 3D Scan Registration for Autonomous Vehicles** Oct'16 - Feb '17  
*Guided by Prof. Gaurav Pandey at IIT Kanpur*

- Designed a fast Mutual Information based novel scan alignment algorithm for autonomous vehicles. Speed and robustness was found to be better than current state of the art scan matching methods when tested on two real time dynamic datasets (containing both rural and urban areas) recorded using a Lidar. Cuda based GPU implementation available. [\[code\]](#)
- Worked on a super-voxel clustering algorithm for 3D point clouds to represent an efficient Gaussian Mixture Model (GMM). This model was then used to apply a point-to-distribution alignment method for registration.

## PUBLICATION

- **Nikhil Mehta**, James R. McBride & Gaurav Pandey, “**Robust and Fast 3D Scan Alignment using Mutual Information**”. International Conference on Robotics and Automation (ICRA) 2018 [\[Accepted\]](#) [\[pdf\]](#)
- **Nikhil Mehta** & Piyush Rai, “**Deep Generative Latent Feature Relational Models**”. [Under Review at ICML 2018]

## INDUSTRY EXPERIENCE

### Senior Project Associate at IIT-Kanpur

August 2016 - Present

- Research Assistant in the Machine Learning group at the Department of Computer Science. Working at the intersection of probabilistic machine learning and deep learning models.
- Research Assistant in the Autonomous Vehicle Group. Worked on applications of Deep Learning and Sensor Fusion for self-driving vehicles.

### Software Engineer at Fitso

Jan - Aug '16

- Was the third employee in the technical team at the startup. Had responsibility of various critical components including back-end and front-end development of the product.
- Had complete responsibility for the android application development. Engineered the product that supported rapid growth from 20,000 users to 100,000 global users in less than six months. Application was selected as the Editors' Choice at Google play store. [\[link\]](#)

### Software Engineer at Zomato

June '14 - Jan '16

- Was a core member of the technical team. Developed server side APIs for various consumer and enterprise products provided by Zomato across 22 countries including India, US and Canada.
- Was part of the mobile development team with only 3 people where we were responsible for the development and release cycles of the primary Zomato application which had more than 5 million users. [\[link\]](#)

## SELECTED PROJECTS & INTERNSHIPS

### Research Intern at University of Birmingham, UK

Dec '13 - Jan '14

Developed a paradigm and an experimental game in Python to investigate how people's perception of a hidden partner affect their behavior while competing against an artificial intelligent bot. Used the existing speech recognition model to create an end-end application for voice based commands.

### Research Intern at University of Leeds, UK

Jul - Aug '13

Designed a vision algorithm for cell (lamallopodia) tracking in DIC microscopic videos using entropy differences in subsequent frames. Real-time implementation was further improved by implementing the Extended Kalman filter.

### Severity prediction using data mining and machine learning

*Final year major project guided by Prof. Ruchika Malhotra at DTU*

Jan - May '14

Worked on document (software documentations) feature extraction using existing methods such as tf-idf, bag-of-words and word2vec. Formulated a multi-class classification problem using an ensemble of various models such as decision trees, support vector machines and naive bayes classifiers for severity prediction.

## TECHNICAL SKILLS

**Deep Learning Frameworks:** Tensorflow, Pytorch, Theano

**Languages:** Python, C, C++, Java, CUDA for GPU, MATLAB, PHP, SQL,  $\text{\LaTeX}$ .

**Statistical Modeling:** Stan, Edward (Bayesian methods in deep networks).

## RELEVANT COURSEWORK

**Computer Science:** Data Structures, Algorithms, Artificial Intelligence, Compilers, Theory of Computation, Operating Systems

**Mathematics:** Convex Optimization, Linear Algebra, Probabilistic Models

**MOOCs:** Machine Learning, Neural Networks for Machine Learning, Parallel GPU Programming using CUDA, Game Development [\[Certifications\]](#)

## TEACHING EXPERIENCE

- **Project Mentor, Probabilistic Machine Learning, IIT Kanpur**

Course Instructor: Prof. Piyush Rai

- **Technical Head, Society of Software Engineering, DTU**

Competitive Programming Mentor, Chief Coordinator: Prof. Ruchika Malhotra